Blockchain-Based Academic Degree Verification & Credential Authentication

BLOCKCHAIN & CYBERSECURITY

SOFTWARE PROJECT

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The Problem: Academic Credential Fraud Is Rampant

Fake Certificates

Forged degrees and manipulated transcripts are widespread, undermining trust in academic qualifications across industries.

Slow & Manual Verification

Traditional verification relies on physical documents and direct university communication — a process that is slow, expensive, and error-prone.

No Decentralized Standard

Employers and institutions lack a secure, transparent, and universally accessible system to verify credentials quickly and reliably.

Our Proposed Solution

We propose a Blockchain-Based Degree Verification System where:

- Universities issue certificates digitally
- Certificate proof is stored securely on blockchain
- Students can access and share verified credentials
- Employers can instantly verify certificates using QR codes or certificate IDs
- Smart contracts handle issuance, validation, and revocation

Project Objectives

The main objectives are:

- Eliminate fake degree fraud
- Reduce certificate verification time
- Build trust between universities and employers
- Create permanent tamper-proof academic records
- Enable instant global verification
- Support digital education transformation
- Reduce administrative workload for institutions

Technology Used

Frontend:
React.js / Next.js

Backend:
Node.js / Express.js

Blockchain:
Ethereum / Polygon / Hyperledger

Smart Contracts:
Solidity

Database:
MongoDB / PostgreSQL

Innovation: AI + Blockchain Hybrid Verification

Beyond standard blockchain verification, our system integrates **Artificial Intelligence** for proactive fraud detection — making it truly hackathon-worthy.

Duplicate Detection

Identifies cloned or duplicated certificates across the network.

Pattern Recognition

Detects fake university naming patterns and suspicious credential structures.

CGPA Anomaly Analysis

Flags statistically improbable grade manipulations using ML models.

Predictive Prevention

AI models predict and prevent fraud before certificates are issued.

Future Scope & Roadmap

1

Phase 1 — DigiLocker Integration

Connect with India's national digital document wallet for seamless citizen access.

2

Phase 2 — Government Board Support

Onboard state and central education boards for standardized credential issuance.

3

Phase 3 — International Universities

Expand verification network to global institutions for cross-border credential recognition.

4

Phase 4 — NFT Credentials & Mobile App

Issue credentials as soulbound NFTs and launch a dedicated mobile application.

Conclusion

By combining **Blockchain, Smart Contracts, IPFS, and AI**, this system delivers a practical, scalable, and secure solution to one of education's most pressing real-world challenges.

Real-World Impact

Solves a national-scale problem affecting millions of students, employers, and institutions.

Hackathon-Ready

Innovative, technically robust, and highly relevant for Smart India Hackathon and final-year projects.

Production-Grade

Built with enterprise technologies designed for real deployment, not just demonstration.

THANK YOU!